

UNIT 1- INTRODUCTION

Definition:

Statistics is a branch of mathematics dealing with the collection, analysis, interpretation, presentation, and organization of data. In applying statistics to, e.g., a scientific, industrial, or social problem, it is conventional to begin with a statistical population or a statistical model process to be studied. Statistics is also used in associated tasks such as designing experiments and surveys and planning the collection and analysis of data from these.

Seligman explored that statistics is a science that deals with the methods of collecting, classifying, presenting, comparing and interpreting numerical data collected to throw some light on any sphere of enquiry. Spiegel defines statistics highlighting its role in decision-making particularly under uncertainty, as follows: statistics is concerned with scientific method for collecting, organising, summa rising, presenting and analyzing data as well as drawing valid conclusions and making reasonable decisions on the basis of such analysis.

Importance of Statistics:

Statistics plays a vital role in every field of human activity. Statistics helps in determining the existing position of per capita income, unemployment, population growth rates, housing, schooling medical facilities, etc., in a country.

Now statistics holds a central position in almost every field, including industry, commerce, trade, physics, chemistry, economics, mathematics, biology, botany, psychology, astronomy, etc., so the application of statistics is very wide. Now we shall discuss some important fields in which statistics is commonly applied.

(1) Business

Statistics plays an important role in business. A successful businessman must be very quick and accurate in decision making. He knows what his customers want; he should therefore know what to produce and sell and in what quantities.

Statistics helps businessmen to plan production according to the taste of the customers, and the quality of the products can also be checked more efficiently by using statistical methods. Thus, it can be seen that all business activities are based on statistical information. Businessmen can make correct decisions about the location of business, marketing of the products, financial resources, etc.

(2) Economics

Economics largely depends upon statistics. National income accounts are multipurpose indicators for economists and administrators, and statistical methods are used to prepare these accounts. In economics research, statistical methods are used to collect and analyze the data and test hypotheses. The relationship between supply and demand is studied by statistical methods; imports and exports, inflation rates, and per capita income are problems which require a good knowledge of statistics.

(3) Mathematics

Statistics plays a central role in almost all natural and social sciences. The methods used in natural sciences are the most reliable but conclusions drawn from them are only probable because they are based on incomplete evidence.

Statistics helps in describing these measurements more precisely. Statistics is a branch of applied mathematics. A large number of statistical methods like probability averages, dispersions, estimation, etc., is used in mathematics, and different techniques of pure mathematics like integration, differentiation and algebra are used in statistics.

(4) Banking

Statistics plays an important role in banking. Banks make use of statistics for a number of purposes. They work on the principle that everyone who deposits their money with the banks does not withdraw it at the same time. The bank earns profits out of these deposits by lending it to others on interest. Bankers use statistical approaches based on probability to estimate the number of deposits and their claims for a certain day.

(5) State Management (Administration)

Statistics is essential to a country. Different governmental policies are based on statistics. Statistical data are now widely used in making all administrative decisions. Suppose if the government wants to revise the pay scales of employees in view of an increase in the cost of living, and statistical methods will be used to determine the rise in the cost of living. The preparation of federal and provincial government budgets mainly depends upon statistics because it helps in estimating the expected expenditures and revenue from different sources. So statistics are the eyes of the administration of the state.

(6) Accounting and Auditing

Accounting is impossible without exactness. But for decision making purposes, so much precision is not essential; the decision may be made on the basis of

approximation, known as statistics. The correction of the values of current assets is made on the basis of the purchasing power of money or its current value.

In auditing, sampling techniques are commonly used. An auditor determines the sample size to be audited on the basis of error.

(7) Natural and Social Sciences

Statistics plays a vital role in almost all the natural and social sciences. Statistical methods are commonly used for analyzing experiments results, and testing their significance in biology, physics, chemistry, mathematics, meteorology, research, chambers of commerce, sociology, business, public administration, communications and information technology, etc.

(8) Astronomy

Astronomy is one of the oldest branches of statistical study; it deals with the measurement of distance, and sizes, masses and densities of heavenly bodies by means of observations. During these measurements errors are unavoidable, so the most probable measurements are found by using statistical methods.

Example: This distance of the moon from the earth is measured. Since history, astronomers have been using statistical methods like method of least squares to find the movements of stars.

Presentation of Data:

Once data has been collected, it has to be classified and organised in such a way that it becomes easily readable and interpretable, that is, converted to information. Before the calculation of descriptive statistics, it is sometimes a good idea to present data as tables, charts, diagrams or graphs. Most people find 'pictures' much more helpful than 'numbers' in the sense that, in their opinion, they present data more meaningfully.

Data can be classified as grouped or ungrouped.

Ungrouped data are data that are not organized, or if arranged, could only be from highest to lowest or lowest to highest. Grouped data are data that are organized and arranged into different classes or categories.

Presentation of Data:

Textual Method	Tabular Method	Graphical Method
----------------	----------------	------------------

Rearrangement from lowest to highest • Stem-and-leaf plot	•Frequency distribution table (FDT) •Relative FDT •Cumulative FDT • Contingency Table	• Bar Chart •Histogram •Frequency Polygon • Pie Chart •Less than, greater than Ogive
--	--	--

Measures of Central Tendency:

The description of statistical data may be quite elaborate or quite brief depending on two factors: the nature of data and the purpose for which the same data have been collected. While describing data statistically or verbally, one must ensure that the description is neither too brief nor too lengthy. The measures of central tendency enable us to compare two or more distributions pertaining to the same time period or within the same distribution over time. For example, the average consumption of tea in two different territories for the same period or in a territory for two years, say, 2003 and 2004, can be attempted by means of an average.

The central tendency of a variable means a typical value around which other values tend to concentrate; hence this value representing the central tendency of the series is called measures of central tendency or average.

The following are the five measures of average or central tendency that are in common use : (i) Arithmetic mean or simple mean (ii) Median (iii) Mode (iv) Geometric mean (v) Harmonic mean.

Arithmetic mean, Geometric mean and Harmonic means are usually called Mathematical averages while Mode and Median are called Positional averages.

Characteristics for a good or an ideal average :

The following properties should possess for an ideal average.

It should be rigidly defined.

It should be easy to understand and compute.

It should be based on all items in the data.

Its definition shall be in the form of a mathematical formula.

It should be capable of further algebraic treatment.

It should have sampling stability.

It should be capable of being used in further statistical computations or processing.

Besides the above requisites, a good average should represent maximum characteristics of the data, its value should be nearest to the most items of the given series

Arithmetic mean:

To find the arithmetic mean, add the values of all terms and then divide sum by the number of terms, the quotient is the arithmetic mean. The most popular and widely used measure of representing the entire data by one value is known as arithmetic mean. Its value is obtained by adding together all the items and by dividing this total by the number of items.

Individual Series	Discrete & continuous series
Direct Method : $X = \frac{\sum X}{N}$ Where – X □ Values N □ No. of Items	Direct Method : $X = \frac{\sum fX}{N}$ Where – fX □ Values X frequencies N □ Total of frequencies
Shortcut Method : $X = A + \frac{\sum dx}{N}$ Where – A □ Assumed Mean dx □ $X - A$	Shortcut Method : $\bar{X} = A + \frac{\sum f dx}{N}$

Step-Deviation Method : Not applicable	Step-Deviation Method : $\bar{X} = A + \frac{\sum f dx}{N}$ Where – $dx = X_i - A$ $i = \text{Class interval}$
--	---

Merits and demerits of Arithmetic mean :

Merits:

It is rigidly defined.

It is easy to understand and easy to calculate.

If the number of items is sufficiently large, it is more accurate and more reliable.

It is a calculated value and is not based on its position in the series.

It is possible to calculate even if some of the details of the data are lacking.

Of all averages, it is affected least by fluctuations of sampling.

It provides a good basis for comparison.

Demerits:

It cannot be obtained by inspection nor located through a frequency graph.

It cannot be in the study of qualitative phenomena not capable of numerical measurement i.e. Intelligence, beauty, honesty etc.,

It can ignore any single item only at the risk of losing its accuracy.

It is affected very much by extreme values.

It cannot be calculated for open-end classes.

It may lead to fallacious conclusions, if the details of the data from which it is computed are not given.

Weighted Arithmetic mean :

For calculating simple mean, we suppose that all the values or the sizes of items in the distribution have equal importance. But, in practical life this may not be so. In case some items are more important than others, a simple

average computed is not representative of the distribution. Proper weightage has to be given to the various items. For example, to have an idea of the change in cost of living of a certain group of persons, the simple average of the prices of the commodities consumed by them will not do because all the commodities are not equally important, e.g rice, wheat and pulses are more important than tea, confectionery etc., It is the weighted arithmetic average which helps in finding out the average value of the series after giving proper weight to each group.

Definition:

The average whose component items are being multiplied by certain values known as “weights” and the aggregate of the multiplied results are being divided by the total sum of their “weight”.

If $x_1, x_2 \dots x_n$ be the values of a variable x with respective weights of $w_1, w_2 \dots w_n$ assigned to them, then

$$\text{Weighted A.M } \bar{x}_w = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + \dots + w_n} = \frac{\sum w_i x_i}{\sum w}$$

Uses of the weighted mean:

Weighted arithmetic mean is used in:

- Construction of index numbers.
- Comparison of results of two or more universities where number of students differ.
- Computation of standardized death and birth rates.

Geometric Mean

Geometric mean is the n^{th} root of the product of N items or values

The geometric mean of a series containing n observations is the n^{th} root of the product of the values. If $x_1, x_2 \dots, x_n$ are observations then

$$\log GM = \frac{1}{n} \log(x_1 \cdot x_2 \dots x_n)$$

$$\begin{aligned}
 & \frac{1}{n} (\log x_1 + \log x_2 + \dots + \log x_n) \\
 &= \frac{\sum \log x_i}{n} \\
 &= \frac{\log \left(\frac{x_1 \cdot x_2 \cdot \dots \cdot x_n}{n} \right)}{n}
 \end{aligned}$$

Individual Series	Discrete & Continuous Series
$G = \text{Antilog} \left(\frac{\sum \log X}{N} \right)$	$G = \text{Antilog} \left(\frac{\sum f \log X}{N} \right)$

Merits of Geometric mean :

It is rigidly defined

It is based on all items

It is very suitable for averaging ratios, rates and percentages

It is capable of further mathematical treatment.

Unlike AM, it is not affected much by the presence of extreme values

Demerits of Geometric mean:

It cannot be used when the values are negative or if any of the observations is zero

It is difficult to calculate particularly when the items are very large or when there is a frequency distribution.

It brings out the property of the ratio of the change and not the absolute difference of change as the case in arithmetic mean.

The GM may not be the actual value of the series.

Algebraic Characteristics of Geometric Mean :

- (ii) The product of the items remains unchanged if each item is replaced by geometric mean.
- (iii) Geometric mean cannot be found if the value of some item in the series is negative or zero.

- (iv) The product of corresponding ratios on either side of the geometric mean is always equal.
- (v) Not affected by changing the sequence of items. Geometric mean is **appropriate or useful** :-
 - When ratios or percentages are to be found.
 - In determining rates of increase or decrease.
 - When the different values are at vast difference.

Harmonic mean (H.M) :

Harmonic mean of a set of observations is defined as the reciprocal of the arithmetic average of the reciprocal of the given values. If $x_1, x_2, \dots, x_n$ are n observations.

$$H.M = \frac{N}{\sum \frac{f}{x}}$$

Merits of H.M :

It is rigidly defined.

It is defined on all observations.

It is amenable to further algebraic treatment.

It is the most suitable average when it is desired to give greater weight to smaller observations and less weight to the larger ones.

Demerits of H.M :

It is not easily understood.

It is difficult to compute.

It is only a summary figure and may not be the actual item in the series

It gives greater importance to small items and is therefore, useful only when small items have to be given greater weightage

Median (M):

Median is that value of the variable which divides the group into two equal parts, one part comprising all values greater than, and the other all values less than the median. The median is that value of the variate which divides the group into two equal parts, one part comprising all values greater, and the other, all values less than median.

Ungrouped or Raw data :

Arrange the given values in the increasing or decreasing order. If the number of values are odd, median is the middle value. If the number of values are even, median is the mean of middle two values.

By formula

$$\text{Median} = M_d = \frac{n + 1}{2}^{\text{th}} \text{ item.}$$

Grouped Data:

In a grouped distribution, values are associated with frequencies. Grouping can be in the form of a discrete frequency distribution or a continuous frequency distribution. Whatever may be the type of distribution, cumulative frequencies have to be calculated to know the total number of items.

Cumulative frequency : (cf)

Cumulative frequency of each class is the sum of the frequency of the class and the frequencies of the previous classes, ie adding the frequencies successively, so that the last cumulative frequency gives the total number of items.

Continuous Series:

$$M = L_1 + \frac{N/2 - cf}{f} * c$$

L_1 □ lower limit of median group

c □ class interval

cf □ cumulative frequency before median class.

f □ frequency of median class

Ogive Curve

Suppose we are given the following series:

<i>Class interval</i>	0-10	10-20	20-30	30-40	40-50	50-60	60-70
<i>Frequency</i>	6	12	22	37	17	8	5

We are asked to draw both types of ogive from these data and to determine the

median.

Solution:

First of all, we transform the given data into two cumulative frequency distributions,

one based on 'less than' and another on 'more than' methods.

Table A

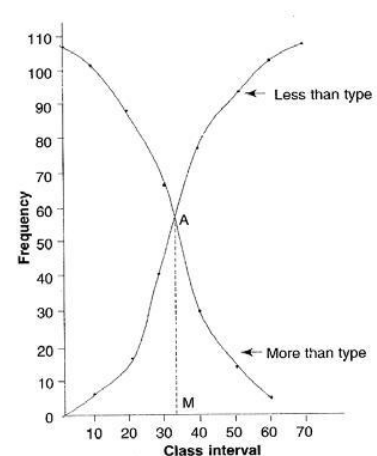
	<i>Frequency</i>
Less than 10	6
Less than 20	18
Less than 30	40
Less than 40	77
Less than 50	94
Less than 60	102
Less than 70	107

Table B

	Frequency
More than 0	107
More than 10	101

More than 20	89
More than 30	67
More than 40	30
More than 50	13
More than 60	5

It may be noted that the point of intersection of the two ogives gives the value of the median. From this point of intersection A, we draw a straight line to



meet the X-axis at M. Thus, from the point of origin to the point at M gives the value of the median, which comes to 34, approximately. If we calculate the median by applying the formula, then the answer comes to 33.8, or 34, approximately. It may be pointed out that even a single ogive can be used to determine the median. As we have determined the median graphically, so also we can find the values of quartiles, deciles or percentiles graphically. For example, to determine we have to take size of $\{3(n + 1)\} / 4 = 81^{\text{st}}$ item. From this point on the Y-axis, we can draw a perpendicular to meet the 'less than' ogive from which another straight line is to be drawn to meet the X-axis. This point will give us the value of the upper quartile. In the same manner, other values of Q_1 and deciles and percentiles can be determined.

Merits of Median :

Median is not influenced by extreme values because it is a positional average.

Median can be calculated in case of distribution with open-end intervals.

Median can be located even if the data are incomplete.

Median can be located even for qualitative factors such as ability, honesty etc.

Demerits of Median :

A slight change in the series may bring drastic change in median value.

In case of even number of items or continuous series, median is an estimated value other than any value in the series.

It is not suitable for further mathematical treatment except its use in mean deviation.

It is not taken into account all the observations.

Quartiles :

The quartiles divide the distribution in four parts. There are three quartiles. The second quartile divides the distribution into two halves and therefore is the same as the median. The first (lower) quartile (Q_1) marks off the first one-fourth, the third (upper) quartile (Q_3) marks off the three-fourth.

Raw or ungrouped data:

First arrange the given data in the increasing order and use the formula for Q_1 and Q_3 then quartile deviation, Q.D is given by

$$Q.D = \frac{Q_3 - Q_1}{2}$$

$$\text{Where } Q_1 = \frac{n+1}{4}^{\text{th}} \text{ item and } Q_3 = \frac{3(n+1)}{4}^{\text{th}} \text{ item}$$

Percentiles :

The percentile values divide the distribution into 100 parts each containing 1 percent of the cases. The percentile (P_k) is that value of the variable up to which lie exactly $k\%$ of the total number of observations.

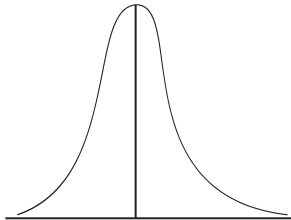
Mode:

The mode refers to that value in a distribution, which occur most frequently. It is an actual value, which has the highest concentration of items in and around it.

According to Croxton and Cowden “ The mode of a distribution is the value at the point around which the items tend to be most heavily concentrated. It may be regarded at the most typical of a series of values”.

It shows the centre of concentration of the frequency in around a given value. Therefore, where the purpose is to know the point of the highest concentration it is preferred. It is, thus, a positional measure.

Its importance is very great in marketing studies where a manager is interested in knowing about the size, which has the highest concentration of items. For example, in placing an order for shoes or ready-made garments the modal size helps because this sizes and other sizes around in common demand.



Determination of Modal class by grouping method:

For a frequency distribution modal class corresponds to the maximum frequency. But in any one (or more) of the following cases:

If the maximum frequency is repeated

If the maximum frequency occurs in the beginning or at the end of the distribution

If frequency is concentrated

If there are irregularities in the distribution, the modal class is determined by the method of grouping.

Steps for Calculation :

We prepare a grouping table with 6 columns

In column I, we write down the given frequencies.

Column II is obtained by combining the frequencies two by two.

Leave the 1st frequency and combine the remaining frequencies two by two and write in column III

Column IV is obtained by combining the frequencies three by three.

Leave the 1st frequency and combine the remaining frequencies three by three and write in column V

Leave the 1st and 2nd frequencies and combine the remaining frequencies three by three and write in column VI

Mark the highest frequency in each column. Then form an analysis table to find the modal class. After finding the modal class use the formula to calculate the modal value.

$$\text{Mode} = l_1 + \frac{f_1 - f_0}{(f_1 - f_0) + (f_1 - f_2)} \times i$$

Where, l_1 = the lower value of the class in which the mode lies

f_1 = the frequency of the class in which the mode lies

f_0 = the frequency of the class preceding the modal

class f_2 = the frequency of the class succeeding the

modal class i = the class-interval of the modal class

Merits of Mode:

It is easy to calculate and in some cases it can be located mere inspection

Mode is not at all affected by extreme values.

It can be calculated for open-end classes.

It is usually an actual value of an important part of the series.

In some circumstances it is the best representative of data.

Demerits of mode:

It is not based on all observations.

It is not capable of further mathematical treatment.

Mode is ill-defined generally, it is not possible to find mode in some cases.

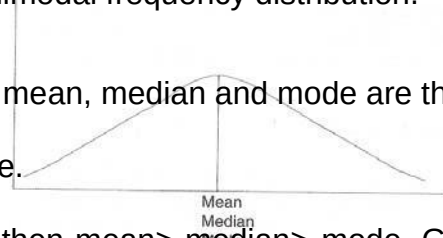
As compared with mean, mode is affected to a great extent, by sampling fluctuations.

It is unsuitable in cases where relative importance of items has to be considered.

RELATIONSHIPS OF THE MEAN, MEDIAN AND MODE

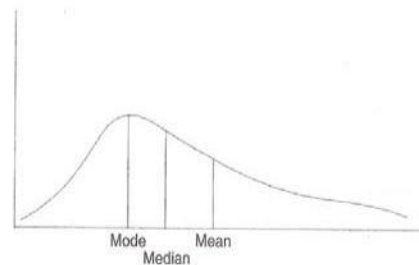
Having discussed mean, median and mode, we now turn to the relationship amongst these three measures of central tendency. We shall discuss the relationship assuming that there is a unimodal frequency distribution.

When a distribution is symmetrical, the mean, median and mode are the same, as is shown below in the following figure.



In case, a distribution is skewed to the right, then $\text{mean} > \text{median} > \text{mode}$. Generally, income distribution is skewed to the right where a large number of families have relatively low income and a small number of families have extremely high income. In such a case, the mean is pulled up by the extreme high incomes and the relation among these three measures is as shown in Fig. 6.3. Here, we find that $\text{mean} > \text{median} > \text{mode}$.

When a distribution is skewed to the left, then $\text{mode} > \text{median} > \text{mean}$. This is because here mean is pulled down below the median by extremely low values. This is



shown as in the figure.

Given the mean and median of a unimodal distribution, we can determine whether it is skewed to the right or left. When $\text{mean} > \text{median}$, it is skewed to the right; when $\text{median} > \text{mean}$, it is skewed to the left. It may be noted that the median is always in the middle between mean and mode.

THE BEST MEASURE OF CENTRAL TENDENCY

At this stage, one may ask as to which of these three measures of central tendency the best is. There is no simple answer to this question. It is because these three measures are based upon different concepts. The arithmetic mean is the sum of the values divided by the total number of observations in the series. The median is the value of the middle observation that divides the series into two equal parts. Mode is the value around which the observations tend to concentrate. As such, the use of a particular measure will largely depend on the purpose of the study and the nature of the data; For example, when we are interested in knowing the consumers preferences for different brands of television sets or different kinds of advertising, the choice should go in favour of mode. The use of mean and median would not be proper. However, the median can sometimes be used in the case of qualitative data when such data can be arranged in an ascending or descending order. Let us take another example. Suppose we invite applications for a certain vacancy in our company. A large number of candidates apply for that post. We are now interested to know as to which age or age group has the largest concentration of applicants. Here, obviously the mode will be the most appropriate choice. The arithmetic mean may not be appropriate as it may be influenced by some extreme values. However, the mean happens to be the most commonly used measure of central tendency as will be evident from the discussion in the subsequent chapters.

What do you mean by Dispersion. Give the meaning of Absolute Measure and Relative Measure with example.

Dispersion is a measure of the extent to which the individual item vary from a central value Dispersion is used in two senses, (i) difference between the extreme items of the series and (ii) average of deviation of items from the mean.

Absolute Measure : The figure showing the limit or magnitude of dispersion is known as absolute measure and it is shown in the same unit as ;those of the original data, example measures of dispersion in the

age of students, their height, weight etc.

Relative Measure : For comparative study the concerning absolute measure is divided by the corresponding mean or some other characteristic value to obtain a ratio or percentage, which is known as the relative measure.

The various methods absolute and relative measures of dispersion are listed below

Absolute measure	Relative measure
1. Range	1. Co-efficient of Range
	Co-efficient of Quartile
2. Quartile deviation	2. deviation
3. Mean deviation	3. Co-efficient of Mean deviation
4. Standard deviation	4. Co-efficient of variation

i) **Range**: The difference between the value of the smallest item and the value of the largest item.
Range = Largest item – Smallest item $L - S$.

Where L = Largest value.

S = Smallest value

$$\text{Co-efficient of Range} = \frac{L - S}{L + S}$$

Merits of Range :

- Simple and easy to be computed.
- It takes minimum time to calculate.
- Not necessary to know all the values, only smallest and largest value is required.
- Helpful in quality control of products.

Demerits of Range :

- Not based on all the items.
- Subject to fluctuation/uncertain measure.
- Cannot be computed in case of open-end distributions.

As it is not based on all the values it is not considered as a good or appropriate measure.

Quartile Deviation

Quartile Deviation gives the average amount by which the two quartiles differ from the median. Quartile deviation is an absolute measure of dispersion.

It is calculated by using the formula -

$$\text{Q.D.} = \frac{Q_3 - Q_1}{2}, \text{ coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

Merits and Demerits of Quartile Deviation

Merits :

It is Simple to understand and easy to calculate

It is not affected by extreme values.

It can be calculated for data with open end classes also.

Demerits:

It is not based on all the items. It is based on two positional values Q_1 and Q_3 and ignores the extreme 50% of the items

It is not amenable to further mathematical treatment.

It is affected by sampling fluctuations

Mean Deviation:

Individual Series:

$$\text{M.D(about mean)} = \frac{\sum |X - \bar{X}|}{N}$$

$$\text{M.D(about median)} = \frac{\sum |X - \text{Median}|}{N}$$

$$\text{M.D(about mode)} = \frac{\sum |X - \text{Mode}|}{N}$$

Continuous Series

$$\text{M.D(about mean)} = \frac{\sum f|X - \bar{X}|}{\sum f}$$

$$\text{M.D(about median)} = \frac{\sum f|X - \text{Median}|}{\sum f}$$

$$\text{M.D(about mode)} = \frac{\sum f|X - \text{Mode}|}{\sum f}$$

$\sum$, represents the summation

X , represents the observation.

$\bar{X}$ represents the mean.

N represents the number of observation.

Coefficient of Mean Deviation:

$$\frac{\text{M.D(about mean)}}{\bar{X}}$$

$$\frac{\text{M.D(about median)}}{\text{Median}}$$

$$\frac{\text{M.D(about mode)}}{\text{Mode}}$$

Merits :

- It is simple to understand and easy to compute.
- Based on each and every item of the data.
- Least affected by the extreme values.
- It can be computed from any average, mean, median or mode.

Demerits :

- Signs are ignored therefore mathematically it is incorrect and not a significant measure.
- Cannot be compared if mean deviations of different series are based on different averages.
- Does not give accurate results.

Standard Deviation (σ) :

Standard Deviation was introduced by Karl Pearson in 1823. It is the most important and widely used measure of studying dispersion, as it is free from those defects from which the earlier methods suffer and satisfies most of the properties of a good measure of dispersion.

Standard Deviation is the square root of the average of the square deviations from the arithmetic mean of a distribution.

Individual Series	Discrete & Continuous Series
Direct Method : $\sigma = \sqrt{\frac{\sum d^2}{N}}$ Where – $d^2 = (X - \bar{X})^2$ N = No. of Items	Direct Method : $\sigma = \sqrt{\frac{\sum fd^2}{N}}$
Shortcut Method : $\sigma = \sqrt{\frac{\sum dx^2}{N} - \left(\frac{\sum dx}{N} \right)^2}$ Where – $dx^2 = (X - A)^2$ A = Assumed Mean	Shortcut Method : $\sigma = \sqrt{\frac{\sum fdx^2}{N} - \left(\frac{\sum fdx}{N} \right)^2}$

Coefficient of Variation:

The Standard deviation is an absolute measure of dispersion. It is expressed in terms of units in which the original figures are collected and stated. The standard deviation of heights of students cannot be compared with the standard deviation of weights of students, as both are expressed in different units, i.e heights in centimeter and weights in kilograms. Therefore the standard deviation must be converted into a relative measure of dispersion for the purpose of comparison. The relative measure is known as the coefficient of variation.

The coefficient of variation is obtained by dividing the standard deviation by the mean and multiply it by 100. symbolically,

$$(C.V) = \frac{\sigma}{\bar{X}} \times 100$$

Coefficient of Standard Deviation:

$$= \frac{\sigma}{\bar{X}}$$

$$\text{Variance} = \sigma^2$$

Merits and Demerits of Standard Deviation:

Merits:

It is rigidly defined and its value is always definite and based on all the observations and the actual signs of deviations are used.

As it is based on arithmetic mean, it has all the merits of arithmetic mean.

It is the most important and widely used measure of dispersion.

It is possible for further algebraic treatment.

It is less affected by the fluctuations of sampling and hence stable.

It is the basis for measuring the coefficient of correlation and sampling.

Demerits:

It is not easy to understand and it is difficult to calculate.

It gives more weight to extreme values because the values are squared up.

As it is an absolute measure of variability, it cannot be used for the purpose of comparison.

Skewness:

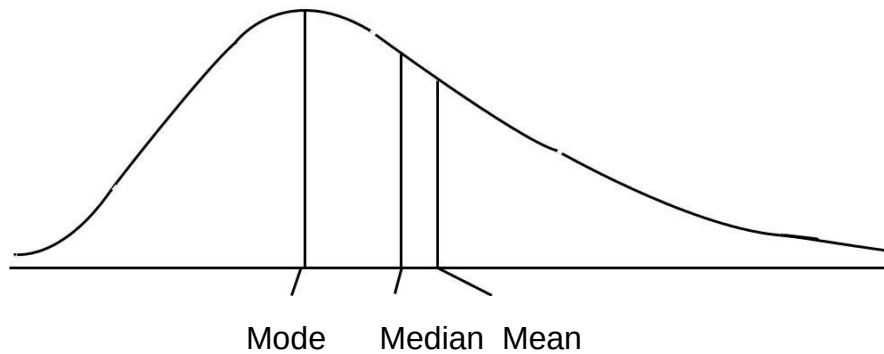
Skewness means 'lack of symmetry'. We study skewness to have an idea about the shape of the curve which we can draw with the help of the given data. If in a distribution mean = median = mode, then that distribution is known as symmetrical distribution. If in a distribution mean $\neq$ median $\neq$ mode, then it is not a symmetrical distribution and it is called a skewed distribution and such a distribution could either be positively skewed or negatively skewed.

a) Symmetrical distribution:

$$\text{Mean} = \text{Median} = \text{Mode}$$

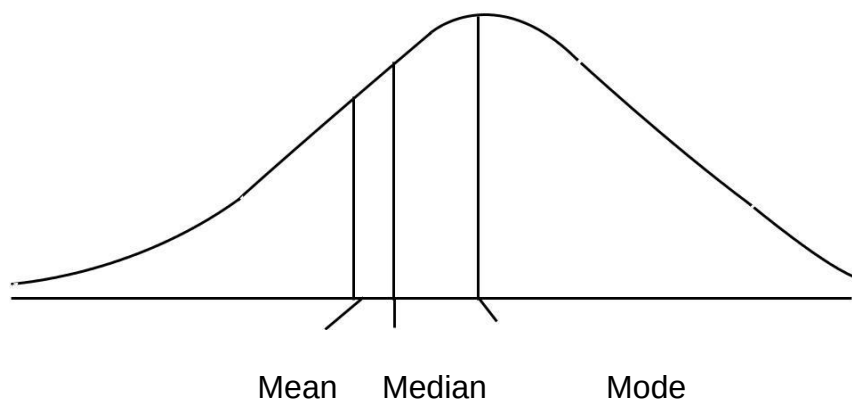
It is clear from the above diagram that in a symmetrical distribution the values of mean, median and mode coincide. The spread of the frequencies is the same on both sides of the center point of the curve.

b) Positively skewed distribution:



It is clear from the above diagram, in a positively skewed distribution, the value of the mean is maximum and that of the mode is least, the median lies in between the two. In the positively skewed distribution the frequencies are spread out over a greater range of values on the right hand side than they are on the left hand side.

c) Negatively skewed distribution:



It is clear from the above diagram, in a negatively skewed distribution, the value of the mode is maximum and that of the mean is least. The median lies in between the two. In the negatively skewed distribution the frequencies are spread out over a greater range of values on the left hand side than they are on the right hand side.

Measures of skewness:

The important measures of skewness are

Karl – Pearson' s coefficient of skewness

Bowley' s coefficient of skewness

Karl – Pearson' s Coefficient of skewness:

According to Karl – Pearson, the absolute measure of skewness = mean – mode. This measure is not suitable for making valid comparison of the skewness in two or more distributions because the unit of measurement may be different in different series. To avoid this difficulty use relative measure of skewness called Karl – Pearson' s coefficient of skewness given by:

$$\text{Karl – Pearson' s Coefficient Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{S.D.}}$$

In case of mode is ill – defined, the coefficient can be determined by the formula:

$$\text{Coefficient of Skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{S.D.}}$$

Bowley' s Coefficient of skewness:

In Karl – Pearson' s method of measuring skewness the whole of the series is needed. Prof. Bowley has suggested a formula based on relative position of quartiles. In a symmetrical distribution, the quartiles are equidistant from the value of the median; ie.,

Median – Q₁ = Q₃ – Median. But in a skewed distribution, the quartiles will not be equidistant from the median. Hence Bowley has suggested the following formula:

$$\text{Bowley' s Coefficient of skewness (sk)} = \frac{Q_3 + Q_1 - 2 \text{ Median}}{Q_3 - Q_1}$$

What is the best method of measuring Dispersion. Write the formula for calculating combined S.D.

Standard Deviation is the best method of measuring dispersion as deviations taken from mean and algebraic signs are not ignored and it is algebraically correct.

Formula for calculating combined Standard Deviation :

$$\sigma_{12} = \sqrt{\frac{N_1 (\sigma_1^2 + D_1^2) + N_2 (\sigma_2^2 + D_2^2) + \dots + N_n (\sigma_n^2 + D_n^2)}{N_1 + N_2 + \dots + N_n}}$$

Where –

N_1 & N_2 = No. of items of two series

σ_1 & σ_2 = S.D. of the two series

$D_1 = \bar{X}_1 - \bar{X}_{12}$

$D_2 = \bar{X}_2 - \bar{X}_{12}$

$\bar{X}_{12}$ = Combined arithmetic mean